VANISHING OF MOD P COHOMOLOGY OF A SEPARATED VARIETY IN CHARACTERISTIC P ABOVE THE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the proper case and the compactification step are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Mod p cohomology of schemes of characteristic p and its vanishing above a dimension bound. Same chapter, the companion vanishing result for quasi-compact and quasi-separated schemes of characteristic p giving vanishing above 2 + d. Supplies the input for the affine and quasi-projective pieces.
- Compactification of a separated scheme of finite type over a field and the associated spectral sequence. Same chapter together with the chapter More on Morphisms, the passage from a separated scheme of finite type to a proper scheme by compactification, together with the spectral sequence computing the cohomology of an open subscheme from that of the compactification and of the boundary. Supplies the reduction from the proper case to the separated case.

Source reference. Original lemma 0A3K: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9612–9612. Navigation pointer invoked at proof line 9847; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0A3M: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9710–9710. Navigation pointer invoked at proof

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line 9850; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0A3L: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9641–9641. Navigation pointer invoked at proof line 9858; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be separated of finite type over an algebraically closed field k of characteristic $p > 0$. Then $H^q_{étale}(X, \underline{\mathbf{Z}/p\mathbf{Z}}) = 0$ for $q \geq dim(X) + 1$.*

Proof. Let $d = \dim(X)$. By the vanishing established in Lemma 0A3K it suffices to show that $H^{d+1}_{étale}(X, \underline{\mathbf{Z}/p\mathbf{Z}}) = 0$. By Lemma 0A3M we see that $H^d(X, \mathcal{O}_X)$ is a finite dimensional k-vector space. Hence the long exact cohomology sequence associated to the Artin-Schreier sequence ends with

$$H^d(X, \mathcal{O}_X) \xrightarrow{F-1} H^d(X, \mathcal{O}_X) \to H^{d+1}_{étale}(X, \mathbf{Z}/p\mathbf{Z}) \to 0$$

By Lemma 0A3L the map $F - 1$ in this sequence is surjective. This proves the lemma. □